Shunchi Zhou

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SUMMARY

B.S.E. software engineer, pursuing master degree of computer science at Ira A. Fulton Schools of Engineering, Arizona State University.

EDUCATION

M.C.S., Computer Science(Big Data Systems) Aug. 2018 - May. 2019
Arizona State University, Tempe, AZ Cumulative GPA: 3.50

B.S.E., Software Engineering Sept. 2013 - Jun.2017
Nantong University, China Major GPA: 3.68

PROJECTS EXPERIENCE

The Implementation and Empirical Research of Vehicle License Plate Recognition Algorithm Based on SVM and ANN
2016.01 - 2016.06

- To develop a system which can detect the vehicles on the road surface and extract the license plate information automatically
- Implementation of machine learning algorithm including using SVM for automatic vehicle license plate detection and ANN to automatically recognize numerical and English characters
- The preliminary analysis of the test results of the whole system

T-LeadLine: Interactive Visual Analysis of Text Data (Top Topic Discovery & Event Exploration)
2018.09 - 2018.12

- A text data-driven event identification and exploration system
- Topic model building, events extracting and summarizing with format of abstraction, spatial-temporal information and related person network, extended with topic ranking feature
- Major techniques including Named-entity Recognition, Latent Dirichlet Allocation, detecting topic burst change (CUSUM), finding time-sensitive terms of given topic and time (TFIDF) and visualizing with D3.js.

Human Activity Recognition from Myo sensors
2019.01 - 2019.05

- Data collection: wear Myo wristband on the right arm for one day, choose one activity from eating, writing, sleeping, and walking, along with the time stamps for such activity
- Feature extraction: select and implement five statistical features (min, max, avg, std, RMS) on each of the time-stamp data
- Dimension Reduction: arrange the feature matrix and run PCA function in MATLAB on the feature matrix

Geo-spatial Hotspot Analysis based on Hadoop and Spark Using Scala language
2019.01 - 2019.05

- Use Apache Hadoop and Apache Spark to implement four basic functions and use them to do spatial queries and hotspot analysis on the dataset of New York City Yellow Cab Taxi Trip Data
- Hot zone analysis: perform a range join operation on a rectangle datasets and a point dataset to obtain the number of points located within the rectangle; calculate the hotness of all the rectangles
- Hot cell analysis: apply spatial statistics to spatio-temporal big data in order to identify statistically significant spatial hot spots using Getis-Ord G_i^* statistic

HONORS

- Scholarships for Students Studying Abroad in the University of Manchester, UK
- Excellent Scholarship(10%)
- Group Award for Undergraduate Service Outsourcing Entrepreneur Competition

TECHNICAL SKILLS

Programming Skills: JAVA, Python, C++, Scala
Database Systems: SQL, PostgreSQL, Hadoop, Spark, MySQL, SQLite
Data Visualization: D3.js
Other: MATLAB, Node.js, Linux, LaTeX